# Tiered Bibliography

## Quantum Mechanics & Quantum Chemistry Programme

*Reading Lists Stratified by Learning Tier*

| | |
|---|---|
| **Tier 1** | High School — Popular Science & First Introductions |
| **Tier 2** | Beginning Undergraduate — Core Textbooks |
| **Tier 3** | Advanced Undergraduate — Rigorous & Bridge Texts |
| **Tier 4** | Master's — Graduate Textbooks & Key Papers |
| **Tier 5** | PhD — Monographs, Reviews & Primary Literature |
| **Cat. A–F** | Full Alphabetical Catalogue (all tiers) |

*Companion document to the Pedagogical Support Material.*

***Annotation key:*** ★★★ *Essential* ★★ *Strongly recommended* ★ *Useful supplement*

***[P]*** *Primary paper* ***[R]*** *Review* ***[M]*** *Monograph* ***[N]*** *Popular/narrative*

# Contents

**How to Use This Bibliography** **2**

**1 Tier 1 (HS) — Popular Science & First Introductions** **3**
1.1 Quantum Mechanics . . . 3
1.2 Nonlinear Dynamics & Chaos (Cross-reference) . . . 4

**2 Tier 2 (BegUG) — Core Undergraduate Textbooks** **5**
2.1 Quantum Mechanics . . . 5
2.2 Mathematics Support . . . 6
2.3 Quantum Chemistry Introduction . . . 6
2.4 Nonlinear Dynamics Cross-reference . . . 6

**3 Tier 3 (AdvUG) — Rigorous Undergraduate & Bridge Texts** **8**
3.1 Quantum Mechanics — Core . . . 8
3.2 Primary Papers — First Encounters . . . 9
3.3 Nonlinear Dynamics Cross-reference . . . 9

**4 Tier 4 (MSc) — Graduate Textbooks & Key Papers** **11**
4.1 Quantum Mechanics — Graduate Texts . . . 11
4.2 Many-Body Theory & Advanced Topics . . . 12
4.3 Quantum Chemistry (Series III) . . . 12
4.4 Key Tier 4 Papers . . . 13

**5 Tier 5 (PhD) — Monographs, Reviews & Primary Literature** **15**
5.1 Functional Analysis & Mathematical Foundations . . . 15
5.2 Advanced Quantum Mechanics . . . 16
5.3 Essential PhD-Level Papers (QM & QC) . . . 17
5.4 Nonlinear Dynamics PhD Cross-reference . . . 17

**6 Full Alphabetical Catalogue** **19**
Category A: Popular Science . . . 19
Category B: Undergraduate Textbooks . . . 19
Category C: Graduate Textbooks . . . 20
Category D: Research Monographs . . . 20
Category E: Primary Research Papers . . . 21
Category F: Review Articles . . . 22
Tier–Category Quick-Index . . . 22

## How to Use This Bibliography

This bibliography is a companion to the **Pedagogical Support Material** (PSM), which describes the five-tier learning system and the 14-type assessment artifact set. Each tier corresponds to a defined competency level; the reading lists here feed directly into the PSM's learning objectives, concept questions, and research tracks.

| Tier | Label | PSM track | Reading strategy |
|---|---|---|---|
| 1 | HS | Undergrad | Popular books cover-to-cover first |
| 2 | BegUG | Undergrad | One textbook chapter per lecture |
| 3 | AdvUG | Undergrad | Textbook + primary paper abstracts |
| 4 | MSc | Graduate | Primary papers as main reading |
| 5 | PhD | Graduate + Research | Monographs + arXiv preprints |

The tiers are **cumulative**: all Tier $k$ reading is recommended as background for Tier $k+1$. Each entry carries:

- A **star rating** (★★★ / ★★ / ★)
- A **type tag** ([P] [R] [M] [N])
- A **short annotation** linking the work to specific programme modules

**Quantum Mechanics** entries are always primary; cross-references to **Nonlinear Dynamics & Chaos** (the logistic map course) are included at each tier to illustrate how the same tier architecture transfers across disciplines.

# 1 Tier 1 (HS) — Popular Science & First Introductions

## Tier 1 — High School (HS)

**Goal:** Build vivid physical intuition and historical awareness before encountering any formalism. All books in this tier require only high-school mathematics. Read these *before* any formal course. They provide the mental pictures that make the formalism of Tiers 2–5 meaningful.

### 1.1 Quantum Mechanics

**Feynman, R. P.**
*QED: The Strange Theory of Light and Matter*
Princeton University Press, 1985.

★★★ **[N]**

*The finest popular account of quantum amplitudes and interference. Feynman derives the Born rule from "arrows" that rotate and add — precisely the Tier 1 picture of the bra-ket formalism. The "photon bouncing off glass" chapters are a perfect pre-reading for Module I.2, Lectures L01–L02. Read before any formal course.*

**Feynman, R. P., Leighton, R. B. & Sands, M.**
*The Feynman Lectures on Physics, Vol. III: Quantum Mechanics*
Addison-Wesley, 1965. Freely available at `feynmanlectures.caltech.edu`

★★★ **[N/pedagogical]**

*Volume III opens with two-slit interference and amplitudes; the first four chapters are fully accessible at Tier 1 and map directly onto the Tier 1 engagement activities for Module I.2 Lectures L01 and L02. An essential free resource.*

**Cox, B. & Forshaw, J.**
*The Quantum Universe: Everything That Can Happen Does Happen*
Da Capo Press, 2011.

★★ **[N]**

*Readable narrative covering superposition, entanglement, and the Pauli exclusion principle with minimal formalism. Good companion for Module I.2 Lectures L09–L10 (tensor products, spin). Can be read in a weekend.*

**Kumar, M.**
*Quantum: Einstein, Bohr, and the Great Debate about the Nature of Reality*
Norton, 2008.

★★ **[N]**

*Historical narrative of the foundational debates about measurement and probability. Essential context for understanding why Born's rule (1926) was controversial. Prepares the historical framing for Module I.2 Lectures L04 and L07.*

**Penrose, R.**
*The Emperor's New Mind*, Chapters 5–6
Oxford University Press, 1989.

★ **[N]**

*Philosophical context for quantum mechanics. Chapters 5–6 discuss wave functions and measurement in an accessible way. Useful philosophical background before Module I.3 Lectures L05 and L07.*

## 1.2 Nonlinear Dynamics & Chaos (Cross-reference)

**Gleick, J.**
*Chaos: Making a New Science*
Penguin Books, 1987.

★★★ **[N]**

*Narrative history of May, Feigenbaum, and the logistic map's role in the chaos revolution. No mathematics required. The exact Tier 1 equivalent of Feynman's QED for nonlinear dynamics. Standard first reading for any dynamical systems course.*

**Stewart, I.**
*Does God Play Dice? The New Mathematics of Chaos*
Blackwell, 1989.

★★ **[N]**

*Gentle narrative; introduces bifurcation and sensitive dependence at high-school level.*

# 2 Tier 2 (BegUG) — Core Undergraduate Textbooks

## Tier 2 — Beginning Undergraduate (BegUG)

**Goal:** Develop working fluency in the notation and methods of quantum mechanics. All books assume first-year linear algebra and calculus. Work through one chapter per lecture, solving end-of-chapter exercises alongside SPSU problem sets from the PSM.

### 2.1 Quantum Mechanics

**Griffiths, D. J. & Schroeter, D. F.**
*Introduction to Quantum Mechanics* (3rd ed.)
Cambridge University Press, 2018.

★★★

*The standard first course. Develops wave mechanics before bra-ket notation. Chapters 1–3 (wavefunction, Schrödinger equation, infinite square well) parallel Module I.3 Lectures L02–L05; Chapter 3 introduces Hilbert space and Dirac notation, paralleling Module I.2 Lectures L01–L05. Excellent problem sets map to SPSU format.*

**Shankar, R.**
*Principles of Quantum Mechanics* (2nd ed.)
Springer, 1994.

★★★

*Begins with a self-contained linear algebra review in Chapter 1 — ideal for students bridging Tier 1 to Tier 2. More algebraic than Griffiths; a natural bridge to Sakurai. Chapter 1 pairs with Module I.1; Chapters 4–9 support Module I.2 Lectures L03–L08.*

**Townsend, J. S.**
*A Modern Approach to Quantum Mechanics* (2nd ed.)
University Science Books, 2012.

★★

*Starts from spin-$\frac{1}{2}$ and Dirac notation from the very first chapter (Sakurai-style). Excellent for building bra-ket fluency early. Module I.2 can use this as the parallel text for Lectures L01–L04 and L10.*

**Binney, J. & Skinner, D.**
*The Physics of Quantum Mechanics*
Oxford University Press, 2014. Freely available at `www-thphys.physics.ox.ac.uk`

★★

*British undergraduate style; uses bra-ket from Chapter 1. Excellent treatment of measurement and operators. Good companion for Module I.2 Lectures L03–L07. Free access makes it especially attractive.*

## 2.2 Mathematics Support

**Strang, G.**
*Introduction to Linear Algebra* (5th ed.)
Wellesley–Cambridge Press, 2016.

★★★

*The prerequisite linear algebra text. Chapters 5–6 (eigenvalues, orthogonality) are directly needed for quantum mechanics. Work through before or alongside Module I.1.*

**Byron, F. W. & Fuller, R. W.**
*Mathematics of Classical and Quantum Physics*
Dover, 1992.

★★

*Covers Hilbert spaces, operators, and distributions at a level accessible to advanced Tier 2 students. A useful mathematical supplement for Module I.2 Lectures L06 and L12.*

## 2.3 Quantum Chemistry Introduction

**Jensen, F.**
*Introduction to Computational Chemistry* (3rd ed.)
Wiley, 2017.

★★

*Accessible entry to computational quantum chemistry. Chapters 1–3 provide the Tier 2 background for Series III. Students should read alongside Module III.1.*

## 2.4 Nonlinear Dynamics Cross-reference

**Strogatz, S. H.**
*Nonlinear Dynamics and Chaos* (2nd ed.)
Westview Press, 2015.

★★★

*The standard first course in nonlinear dynamics. Chapters 10–12 cover 1D maps, period-doubling, and chaos with clear physical motivation. Exact Tier 2 analogue of Griffiths.*

**Alligood, K. T., Sauer, T. D. & Yorke, J. A.**
*Chaos: An Introduction to Dynamical Systems*
Springer, 1996.

★★

*Chapter 1 develops the logistic map from scratch with full proofs accessible to beginning undergraduates.*

# 3 Tier 3 (AdvUG) — Rigorous Undergraduate & Bridge Texts

## Tier 3 — Advanced Undergraduate (AdvUG)

**Goal:** Understand why the formalism is internally consistent and geometric. Assumes confidence with analysis and linear algebra. Start reading primary papers at the abstract and introduction level. Use alongside CPSU problem sets from the PSM.

### 3.1 Quantum Mechanics — Core

**Sakurai, J. J. & Napolitano, J.**
*Modern Quantum Mechanics* (3rd ed.)
Cambridge University Press, 2020.

★★★

*The primary text for Module I.2. Begins immediately with bra-ket notation and the Stern–Gerlach experiment. Chapters 1–3 (state vectors, dynamics, angular momentum) are the core of Series I. The Sakurai "operational" approach is one of the two tracks in the Module I.2 dual pedagogy (Sakurai + Dirac). Essential reading at Tier 3.*

**Dirac, P. A. M.**
*The Principles of Quantum Mechanics* (4th ed.)
Oxford University Press, 1958 (reprinted 1981).

★★★ **[P/pedagogical]**

*The original source for bra-ket notation. Not a modern textbook but an indispensable primary reference. Chapter I ("The Principle of Superposition") and Chapter II ("Dynamical Variables and Observables") should be read in full alongside Module I.2. This is the "Dirac track" of the dual pedagogy.*

**Isham, C. J.**
*Lectures on Quantum Theory: Mathematical and Structural Foundations*
Imperial College Press, 1995.

★★

*Bridges undergraduate and graduate levels. Rigorous treatment of Hilbert spaces, operators, and measurement without full functional analysis. Ideal for Tier 3 students aiming at Tier 4; directly supports Module I.2 Lectures L11–L12.*

**Hannabuss, K.**
*An Introduction to Quantum Theory*
Oxford University Press, 1997.

★★

*Mathematically precise undergraduate text. Good treatment of spectral theory, uncertainty, and density operators without measure theory. Supports Module I.2 Lectures L04, L07, and L11.*

## 3.2 Primary Papers — First Encounters

**Born, M.**
*Zur Quantenmechanik der Stoßvorgänge*
Z. Phys. **37**, 863–867 (1926). English translation in Wheeler & Zurek, *Quantum Theory and Measurement* (Princeton, 1983).

★★ **[P]**

*The Born rule. Two pages; the probability interpretation of $|\psi|^2$. Should be read (in translation) by all Tier 3 students. Directly relevant to Module I.2 Lecture L01 ("why probabilities are squared amplitudes") and the WRQ prompt "How did Born's interpretation change the meaning of the wavefunction?"*

**von Neumann, J.**
*Mathematische Grundlagen der Quantenmechanik*
Springer, 1932. English: *Mathematical Foundations of Quantum Mechanics*, Princeton University Press, 1955.

★★ **[M]**

*The mathematical foundation of quantum mechanics. Chapters I–II (abstract Hilbert space, operators) are accessible at Tier 3; Chapters III–VI require Tier 5 preparation. Supports Module I.2 Lecture L12 and Module I.3 Lecture L07.*

**Wigner, E. P.**
*On unitary representations of the inhomogeneous Lorentz group*
Ann. Math. **40**, 149–204 (1939).

★ **[P]**

*Original source for Wigner's theorem and unitary/antiunitary symmetries. Abstract is accessible at Tier 3; full proof requires Tier 5 preparation. Relevant to Module I.2 Lectures L01 and L03 ("symmetries of physical states").*

## 3.3 Nonlinear Dynamics Cross-reference

**Devaney, R. L.**
*An Introduction to Chaotic Dynamical Systems* (2nd ed.)
Addison-Wesley, 1989.

★★★

*Mathematically precise. Introduces Devaney's three conditions for chaos. Bridge text to graduate level; exact Tier 3 analogue of Sakurai for QM.*

**May, R. M.**
*Simple mathematical models with very complicated dynamics*
Nature **261**, 459–467 (1976).

★★★ **[P]**

*The landmark logistic map paper. Eight pages; entirely accessible at Tier 3. The QM analogue for this tier is Born (1926). Should be read at Tier 3 in the nonlinear dynamics course as the first primary paper encounter.*

# 4 Tier 4 (MSc) — Graduate Textbooks & Key Papers

## Tier 4 — Master's (MSc)

**Goal:** Command the full formalism; reproduce key derivations; situate results in the broader literature. Primary papers become the main reading; textbooks serve as reference. Functional analysis background assumed. Use alongside CPSG, SPJG, and WRQ artifact types from the PSM.

### 4.1 Quantum Mechanics — Graduate Texts

**Weinberg, S.**
*Lectures on Quantum Mechanics* (2nd ed.)
Cambridge University Press, 2015.

★★★

*Covers foundations (postulates, measurement, entanglement), symmetries, and perturbation theory at graduate level. Chapter 3 ("Mathematical Tools") is the cleanest graduate treatment of the bra-ket formalism including rigged Hilbert spaces. Supports Module I.2 Lectures L11–L12 and Module I.3 Lectures L01–L04.*

**Galindo, A. & Pascual, P.**
*Quantum Mechanics* (2 vols.)
Springer, 1990–1991.

★★★

*The most mathematically complete standard graduate text. Volume I develops Hilbert space rigorously; Volume II covers scattering, relativistic QM, and second quantisation. Recommended for Tier 4 students aiming at mathematical physics. Supports all of Series I and II.*

**Ballentine, L. E.**
*Quantum Mechanics: A Modern Development* (2nd ed.)
World Scientific, 2014.

★★

*Statistical interpretation; detailed treatment of measurement, density operators, and the ensemble interpretation. Excellent for the foundational content of Module I.2 Lectures L07 and L11, and Module I.3 Lecture L10.*

**Cohen-Tannoudji, C., Diu, B. & Laloë, F.**
*Quantum Mechanics* (2 vols.)
Wiley, 2019 (new edition).

★★

*Encyclopaedic reference. The "Complements" to each chapter are especially valuable for Tier 4 students who need worked examples alongside theory. Supports Modules II.3 (angular momentum) and II.6 (spectroscopy).*

## 4.2 Many-Body Theory & Advanced Topics

**Fetter, A. L. & Walecka, J. D.**
*Quantum Theory of Many-Particle Systems*
McGraw-Hill, 1971 (Dover reprint 2003).

★★★

*The standard graduate text for Series II Modules II.4–II.5. Second quantisation, Green's functions, and Feynman diagrams developed from scratch. Chapter 1 pairs with Module II.4 Lecture L01.*

**Bruus, H. & Flensberg, K.**
*Introduction to Many-Body Quantum Theory in Condensed Matter Physics*
Oxford University Press, 2004. Freely available at `community.dur.ac.uk`

★★★

*More accessible than Fetter–Walecka. Excellent treatment of second quantisation, diagrammatics, and Green's functions. Primary reading for Module II.4. Free access is a strong advantage.*

**Stefanucci, G. & van Leeuwen, R.**
*Nonequilibrium Many-Body Theory of Quantum Systems*
Cambridge University Press, 2013.

★★★

*The definitive text for Module II.5 (NEGF). Develops the Keldysh formalism rigorously from first principles. Chapters 4–6 are the primary reading for Module II.5 Lectures L02–L06.*

## 4.3 Quantum Chemistry (Series III)

**Szabo, A. & Ostlund, N. S.**
*Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory*
McGraw-Hill, 1989 (Dover reprint 1996).

★★★

*The standard entry-level graduate text for Series III. Covers HF theory, CI, MP2, and basic DFT concisely with full derivations. Chapters 1–3 support Module III.1; Chapters 4–6 support Module III.2 Lectures L05–L11.*

**Helgaker, T., Jørgensen, P. & Olsen, J.**
*Molecular Electronic-Structure Theory*
Wiley, 2000.
★★★ **[M]**

*The definitive graduate reference for Modules III.2–III.3. 900 pages covering basis sets, HF, CC, MCSCF, and response theory. Use as the reference text alongside Module III.2 from Lecture L07 onward.*

**Koch, W. & Holthausen, M. C.**
*A Chemist's Guide to Density Functional Theory* (2nd ed.)
Wiley-VCH, 2001.
★★

*Accessible DFT text for Tier 4 chemistry students. Good companion for Module III.3, Lectures L01–L05.*

## 4.4 Key Tier 4 Papers

**Hohenberg, P. & Kohn, W.**
*Inhomogeneous electron gas*
Phys. Rev. **136**, B864 (1964).
★★★ **[P]**

*Proves the two Hohenberg–Kohn theorems. Primary reading for Module III.3 Lecture L01.*

**Kohn, W. & Sham, L. J.**
*Self-consistent equations including exchange and correlation effects*
Phys. Rev. **140**, A1133 (1965).
★★★ **[P]**

*The Kohn–Sham DFT paper. Primary reading for Module III.3. Nobel lecture (1998) provides accessible context. Essential for any Tier 4 quantum chemistry student.*

**Feigenbaum, M. J.**
*Quantitative universality for a class of nonlinear transformations*
J. Stat. Phys. **19**, 25–52 (1978). **[P]**
★★★

*Original discovery of $\delta \approx 4.6692$ and $\alpha$. Cross-reference for Tier 4 nonlinear dynamics students. The QM programme analogue is Kohn & Sham (1965).*

**Li, T.-Y. & Yorke, J. A.**
*Period three implies chaos*
Am. Math. Monthly **82**, 985–992 (1975). **[P]**

★★★

*Coined the word "chaos". Cross-reference Tier 4 example for illustrating how short landmark papers can reshape a field (analogous to Born 1926 in QM).*

# 5 Tier 5 (PhD) — Monographs, Reviews & Primary Literature

## Tier 5 — PhD

**Goal:** Achieve research-level mastery; read primary literature directly; identify open problems; connect the programme to current research fronts. Primary papers and monographs are the main reading; textbooks serve as reference. Use alongside CPJG, WRQ, and ORQ artifact types from the PSM.

### 5.1 Functional Analysis & Mathematical Foundations

**Reed, M. & Simon, B.**
*Methods of Modern Mathematical Physics* (4 vols.)
Academic Press, 1972–1979.

★★★ [M]

*Vol. I (Functional Analysis): Hilbert spaces, operators, spectral theory. Vol. II (Fourier Analysis, Self-Adjointness): self-adjoint extensions, deficiency indices, Gel'fand triple. The mathematical foundation for Module I.2 Lecture L12. Essential for Tier 5. Every PhD student should own these volumes.*

**Rudin, W.**
*Functional Analysis* (2nd ed.)
McGraw-Hill, 1991.

★★★

*Banach and Hilbert spaces, the spectral theorem for bounded operators. Chapter 12 (unbounded operators) is directly relevant to the self-adjoint extension problem for $\hat{p}$, directly supporting Module I.2 Lecture L12 at PhD level.*

**Blank, J., Exner, P. & Havlíček, M.**
*Hilbert Space Operators in Quantum Physics* (2nd ed.)
Springer, 2008.

★★★

*The most physics-oriented rigorous treatment. Covers spectral measures, rigged Hilbert spaces, and quantum dynamics at the PhD level. Directly supports Module I.2 Lectures L11–L12 and Module I.3 Lectures L04–L08.*

**Gel'fand, I. M. & Vilenkin, N. Ya.**
*Generalized Functions, Vol. 4: Applications of Harmonic Analysis*
Academic Press, 1964.

★★ [M]

*Original source for the rigged Hilbert space (Gel'fand triple) framework. Chapter 1 is the reference for Module I.2 Lecture L12 ("Rigged Hilbert Spaces & Mathematical Foundations") at the PhD level.*

## 5.2 Advanced Quantum Mechanics

**Negele, J. W. & Orland, H.**
*Quantum Many-Particle Systems*
Addison-Wesley, 1988 (Perseus reprint 1998).

★★★

*Path integrals and Green's functions at graduate/research level. Supports Modules II.2 and II.5.*

**Kadanoff, L. P. & Baym, G.**
*Quantum Statistical Mechanics*
Benjamin, 1962 (Perseus reprint 1989).

★★★ [M]

*Original derivation of the Kadanoff–Baym equations. Primary reading for Module II.5 Lecture L05.*

**Martin, R. M., Reining, L. & Ceperley, D.**
*Interacting Electrons: Theory and Computational Approaches*
Cambridge University Press, 2016.

★★★

*The definitive modern text connecting DFT, many-body perturbation theory, and quantum Monte Carlo. Covers all of Series III at research depth.*

**Haag, R.**
*Local Quantum Physics: Fields, Particles, Algebras* (2nd ed.)
Springer, 1996.

★★ [M]

*Algebraic QFT; the deepest mathematical treatment of the measurement problem and locality. Background for Series II Module II.4.*

## 5.3 Essential PhD-Level Papers (QM & QC)

**Stone, M. H.**
*On one-parameter unitary groups in Hilbert space*
Ann. Math. **33**, 643–648 (1932).

★★★ **[P]**

*Stone's theorem on strongly continuous unitary groups. The mathematical basis for $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$. Primary reading for Module I.3 Lecture L04 at PhD level.*

**Gleason, A. M.**
*Measures on the closed subspaces of a Hilbert space*
J. Math. Mech. **6**, 885–893 (1957).

★★★ **[P]**

*Derives the Born rule from measure theory on closed subspaces. The deepest axiomatic justification of $P = |\langle a|\psi\rangle|^2$. Essential for Module I.2 Lecture L07 at PhD level and directly relevant to the ORQ prompt "What minimal operational axioms force complex Hilbert space?"*

**Lindblad, G.**
*On the generators of quantum dynamical semigroups*
Commun. Math. Phys. **48**, 119–130 (1976).

★★★ **[P]**

*Derives the Lindblad master equation from complete positivity. Primary reading for Module I.3 Lecture L10 ("Open Systems and Decoherence").*

**Kraus, K.**
*States, Effects, and Operations: Fundamental Notions of Quantum Theory*
Springer Lecture Notes in Physics **190**, 1983.

★★ **[M]**

*Kraus operators, CPTP maps, and the operational formulation of quantum mechanics. Background for Module I.2 Lecture L11 and Module I.3 Lecture L10.*

## 5.4 Nonlinear Dynamics PhD Cross-reference

**de Melo, W. & van Strien, S.**
*One-Dimensional Dynamics*
Springer, 1993. **[M]**

★★★

*Definitive research monograph on interval maps. Proves ACIMs, renormalisation, Feigenbaum theory. Exact analogue of Reed & Simon for 1D dynamics.*

**Lyubich, M.**
*Almost every real quadratic map is either regular or stochastic*
Ann. Math. **156**, 1–78 (2002). [**P**]

★★

*Resolves the measure-theoretic dichotomy for the logistic family; one of the deepest results in 1D dynamics.*

**Young, L.-S.**
*What are SRB measures and which dynamical systems have them?*
J. Stat. Phys. **108**, 733–754 (2002). [**R**]

★★★

*Definitive survey of physical (SRB) invariant measures; directly analogous to density operator surveys in QM.*

# 6 Full Alphabetical Catalogue

Complete alphabetically ordered catalogue of all references across Tiers 1–5 and the programme syllabus, grouped by category. Each entry carries its tier badge(s) and type tag.

## Category A Popular Science & Historical Narrative


**[A1]** T1 Cox, B. & Forshaw, J. *The Quantum Universe.* Da Capo Press, 2011. [**N**]

**[A2]** T1 Feynman, R. P. *QED: The Strange Theory of Light and Matter.* Princeton University Press, 1985. [**N**] ★★★

**[A3]** T1 Feynman, R. P., Leighton, R. B. & Sands, M. *The Feynman Lectures on Physics, Vol. III.* Addison-Wesley, 1965. [**N**] ★★★

**[A4]** T1 Gleick, J. *Chaos: Making a New Science.* Penguin Books, 1987. [**N**] ★★★

**[A5]** T1 Kumar, M. *Quantum: Einstein, Bohr, and the Great Debate.* Norton, 2008. [**N**]

**[A6]** T1 Penrose, R. *The Emperor's New Mind.* Oxford University Press, 1989. [**N**]

**[A7]** T1 Stewart, I. *Does God Play Dice?* Blackwell, 1989. [**N**]


## Category B Undergraduate Textbooks (Tiers 1–3)


**[B1]** T2 Alligood, K. T., Sauer, T. D. & Yorke, J. A. *Chaos: An Introduction to Dynamical Systems.* Springer, 1996.

**[B2]** T2 Binney, J. & Skinner, D. *The Physics of Quantum Mechanics.* Oxford University Press, 2014.

**[B3]** T2 Byron, F. W. & Fuller, R. W. *Mathematics of Classical and Quantum Physics.* Dover, 1992.

**[B4]** T3 Devaney, R. L. *An Introduction to Chaotic Dynamical Systems* (2nd ed.). Addison-Wesley, 1989.

**[B5]** T2 Griffiths, D. J. & Schroeter, D. F. *Introduction to Quantum Mechanics* (3rd ed.). Cambridge University Press, 2018. ★★★

**[B6]** T3 Hannabuss, K. *An Introduction to Quantum Theory.* Oxford University Press, 1997.

**[B7]** T3 Isham, C. J. *Lectures on Quantum Theory.* Imperial College Press, 1995.

**[B8]** T2 Jensen, F. *Introduction to Computational Chemistry* (3rd ed.). Wiley, 2017.

**[B9]** T2 Shankar, R. *Principles of Quantum Mechanics* (2nd ed.). Springer, 1994. ★★★

**[B10]** T2 Strang, G. *Introduction to Linear Algebra* (5th ed.). Wellesley–Cambridge Press, 2016. ★★★

**[B11]** T2 Strogatz, S. H. *Nonlinear Dynamics and Chaos* (2nd ed.). Westview Press, 2015. ★★★

**[B12]** **T2** Townsend, J. S. *A Modern Approach to Quantum Mechanics* (2nd ed.). University Science Books, 2012.

## Category C Graduate Textbooks (Tiers 4–5)

**[C1]** **T4** Ballentine, L. E. *Quantum Mechanics: A Modern Development* (2nd ed.). World Scientific, 2014.

**[C2]** **T5** Blank, J., Exner, P. & Havlíček, M. *Hilbert Space Operators in Quantum Physics* (2nd ed.). Springer, 2008. ★★★

**[C3]** **T4** Bruus, H. & Flensberg, K. *Introduction to Many-Body Quantum Theory.* Oxford University Press, 2004. ★★★

**[C4]** **T4** Cohen-Tannoudji, C., Diu, B. & Laloë, F. *Quantum Mechanics* (2 vols.). Wiley, 2019.

**[C5]** **T4** Fetter, A. L. & Walecka, J. D. *Quantum Theory of Many-Particle Systems.* Dover, 2003. ★★★

**[C6]** **T4** Galindo, A. & Pascual, P. *Quantum Mechanics* (2 vols.). Springer, 1990–1991. ★★★

**[C7]** **T4** Helgaker, T., Jørgensen, P. & Olsen, J. *Molecular Electronic-Structure Theory.* Wiley, 2000. ★★★

**[C8]** **T5** Kadanoff, L. P. & Baym, G. *Quantum Statistical Mechanics.* Perseus, 1989. [M] ★★★

**[C9]** **T4** Koch, W. & Holthausen, M. C. *A Chemist's Guide to Density Functional Theory* (2nd ed.). Wiley-VCH, 2001.

**[C10]** **T5** Martin, R. M., Reining, L. & Ceperley, D. *Interacting Electrons.* Cambridge University Press, 2016. ★★★

**[C11]** **T5** Negele, J. W. & Orland, H. *Quantum Many-Particle Systems.* Perseus, 1998. ★★★

**[C12]** **T5** Rudin, W. *Functional Analysis* (2nd ed.). McGraw-Hill, 1991. ★★★

**[C13]** **T4** Stefanucci, G. & van Leeuwen, R. *Nonequilibrium Many-Body Theory of Quantum Systems.* Cambridge University Press, 2013. ★★★

**[C14]** **T4** Szabo, A. & Ostlund, N. S. *Modern Quantum Chemistry.* Dover, 1996. ★★★

**[C15]** **T4** Weinberg, S. *Lectures on Quantum Mechanics* (2nd ed.). Cambridge University Press, 2015. ★★★

## Category D Research Monographs (Tier 5 & Research Track)

**[D1]** **T5** Collet, P. & Eckmann, J.-P. *Iterated Maps on the Interval as Dynamical Systems.* Birkhäuser, 1980 (reprint 2009). [M]

**[D2]** **T5** de Melo, W. & van Strien, S. *One-Dimensional Dynamics.* Springer, 1993. [M]

★★★

**[D3]** **T5** Gel'fand, I. M. & Vilenkin, N. Ya. *Generalized Functions, Vol. 4.* Academic Press, 1964. [**M**]

**[D4]** **T5** Haag, R. *Local Quantum Physics* (2nd ed.). Springer, 1996. [**M**]

**[D5]** **T5** Kraus, K. *States, Effects, and Operations.* Springer, 1983. [**M**]

**[D6]** **T5** Reed, M. & Simon, B. *Methods of Modern Mathematical Physics* (4 vols.). Academic Press, 1972–1979. [**M**] ★★★

**[D7]** **T3** von Neumann, J. *Mathematical Foundations of Quantum Mechanics.* Princeton University Press, 1955. [**M**]

## Category E Primary Research Papers

**[E1]** **T3** Born, M. Zur Quantenmechanik der Stoßvorgänge. *Z. Phys.* **37**, 863–867 (1926). [**P**] ★★★

**[E2]** **T5** Eckmann, J.-P. & Ruelle, D. Ergodic theory of chaos and strange attractors. *Rev. Mod. Phys.* **57**, 617–656 (1985). [**R**]

**[E3]** **T4** Feigenbaum, M. J. Quantitative universality for a class of nonlinear transformations. *J. Stat. Phys.* **19**, 25–52 (1978). [**P**] ★★★

**[E4]** **T5** Gleason, A. M. Measures on the closed subspaces of a Hilbert space. *J. Math. Mech.* **6**, 885–893 (1957). [**P**] ★★★

**[E5]** **T4** Hohenberg, P. & Kohn, W. Inhomogeneous electron gas. *Phys. Rev.* **136**, B864 (1964). [**P**] ★★★

**[E6]** **T4** Kohn, W. & Sham, L. J. Self-consistent equations including exchange and correlation. *Phys. Rev.* **140**, A1133 (1965). [**P**] ★★★

**[E7]** **T4** Li, T.-Y. & Yorke, J. A. Period three implies chaos. *Am. Math. Monthly* **82**, 985–992 (1975). [**P**]

**[E8]** **T5** Lindblad, G. On the generators of quantum dynamical semigroups. *Commun. Math. Phys.* **48**, 119–130 (1976). [**P**] ★★★

**[E9]** **T5** Lyubich, M. Almost every real quadratic map is either regular or stochastic. *Ann. Math.* **156**, 1–78 (2002). [**P**]

**[E10]** **T3** May, R. M. Simple mathematical models with very complicated dynamics. *Nature* **261**, 459–467 (1976). [**P**] ★★★

**[E11]** **T5** Stone, M. H. On one-parameter unitary groups in Hilbert space. *Ann. Math.* **33**, 643–648 (1932). [**P**] ★★★

**[E12]** **T3** Wigner, E. P. On unitary representations of the inhomogeneous Lorentz group. *Ann. Math.* **40**, 149–204 (1939). [**P**]

**[E13]** **T5** Young, L.-S. Statistical properties of dynamical systems with some hyperbolicity. *Ann. Math.* **147**, 585–650 (1998). [**P**]

**[E14]** **T5** Young, L.-S. What are SRB measures and which dynamical systems have them? *J. Stat. Phys.* **108**, 733–754 (2002). [**R**] ★★★

## Category F Review Articles & Surveys

**[F1]** **T5** Eckmann, J.-P. Roads to turbulence in dissipative dynamical systems. *Rev. Mod. Phys.* **53**, 643–654 (1981). [**R**]

**[F2]** **T5** Lyubich, M. Forty years of unimodal dynamics. *J. Math. Sci.* **170**, 680–713 (2010). [**R**]

**[F3]** **T5** Viana, M. Stochastic dynamics of deterministic systems. *IMPA Lecture Notes* (1997). [**R**]

## Tier–Category Quick-Index

| Tier | Recommended categories and key items |
|---|---|
| **T1** HS | All of Cat. A; Cat. B: [B3, B5, B11]; no primary papers |
| **T2** BegUG | Cat. B: [B2, B5, B8, B9, B10, B11, B12]; Cat. A as context; Cat. E: [E10] (May 1976) abstract only |
| **T3** AdvUG | Cat. B (all); Cat. C: [C4, C15]; Cat. D: [D7]; Cat. E: [E1, E3, E10, E12] in full |
| **T4** MSc | Cat. C (all); Cat. D: [D3, D6, D7]; Cat. E: [E1, E4, E5, E6, E8, E11, E12] as primary reading; Cat. F: [F1, F2] |
| **T5** PhD | Cat. D (all); Cat. E (complete); Cat. F (all); Cat. C as reference; current arXiv preprints |
| Research track | Cat. D (all); Cat. E (all); Cat. F (all); journal club selections; current arXiv |